

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus II

Semester: Spring

Year : 2026
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Change the Cartesian Integrals $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} x dx dy$ to polar form and integrate the polar Integrals. 5
- b) Find the area of the solid that lies in the first octant and is bounded by the coordinate planes, the cylinder $x^2 + y^2 = 9$ and $y^2 + z^2 = 9$. 5
- c) Find the volume bounded by xy plane, the cylinder $x^2 + y^2 = 1$, $x + y + z = 3$ and the plane. 5
2. a) Solve $y'' - 4xy' + (4x^2 - 2)y = 0$ by power series method solution. 7
- b) Express: $2x^2 - 4x + 2$ in terms of Legendre polynomials. And show that : $(n + 1)P_n(x) = P'_{n+1}(x) - xP'_n(x)$. 8

OR

Derive Rodrigue's formula for Legendre polynomials. Also find the Legendre polynomials of $2x^2 - 4x + 2$.

3. a) Find the Laplace transform of following functions 8
 - i. $f(t) = t \sin \sin 3t \cos \cos 2t$
 - ii. $f(t) = \frac{e^{-t} \sin \sin t}{t}$
- b) Solve the initial value problem by using Laplace transform: 7
$$y'' + 2y' + 2y = 5 \sin t, y(0) = y'(0) = 0$$
4. a) A particle moves on the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$ where t is the time. Find the component of velocity and acceleration at $t = 1$ in the direction of $\vec{i} - 3\vec{j} + 2\vec{k}$. 5
- b) Find the directional derivatives of the functions $\vec{f} = xyz^2$ at $P(1,0,3)$ along $\vec{i} - \vec{j} + \vec{k}$. 5

- c) Show that $\vec{F} = \vec{i} + \vec{j} + x^2y\vec{k}$ is irrotational. Also find the scalar potential function $\varphi(x, y, z)$ such that $F = \nabla\varphi$. 5
5. a) State Green's theorem. Apply it to evaluate the integral 7
 $\oint_c [(y - \sin x)dx + \cos x dy]$ where c is the triangle with vertices $(0, 0), (\frac{\pi}{2}, 0), (\frac{\pi}{2}, 1)$.
- b) Evaluate $\iint \vec{F} \cdot \vec{n} \, ds$ by using Gauss Divergence Theorem where 8
 $\vec{F} = (4x, -2y^2, z^2)$ and the surface bounding the region $x^2 + y^2 = 4, z = 0, z = 3$.

OR

- State Stoke's theorem. Using it evaluate the value of $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (y^3, 0, x^3)$ and C is the boundary of the triangle with vertices $(1, 0, 0), (0, 1, 0)$ and $(0, 0, 1)$.
6. a) Find the Fourier series of $f(x) = x^2$ for $-\pi < x < \pi$ and show 5
that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} \dots = \frac{\pi^2}{6}$.
- b) Find the Fourier sine and cosine series of 5
 $f(x) = e^x$ for $0 \leq x \leq l$
- c) Find the general solution of $u_x + u_y - u = 0$. 5
7. Attempt any two questions. 2×5
- a) Derive one dimensional traffic flow model using conservation law.
- b) Evaluate the integral by changing the order of integration:

$$\int_0^1 \int_x^1 \sin(y^2) dy \, dx$$
- c) Find the inverse Laplace transform of $\log \frac{s(s+a)}{(s+b)}$